

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – III

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AIM: Creating Advanced Visualizations in Tableau.

a. Create stacked and grouped bar charts for comparison

Step A: Creating Stacked and Grouped Bar Charts for Comparison

1. Connecting to the Dataset

1. Launch Tableau.
2. Under the Connect panel, click Text File.
3. Select Practical_3(Dataset).csv and click Open.
4. Click Sheet 1.
5. Check the Price field. If it is not numeric, click its data type icon and select Number (Whole).
6. Right-click Price and select Convert to Measure if required.

2. Building a Stacked Bar Chart

1. Drag Product to the Columns shelf.
2. Drag Price to the Rows shelf.
3. Drag Customer ID to the Color option in the Marks card.

Why we are doing it

A stacked bar chart compares the total price of each product while showing the contribution made by different customers.

Usage

Use a stacked bar chart when you want to compare total sales of products and see each customer's contribution.

3. Converting to a Grouped Bar Chart

1. Keep Price on the Rows shelf.
2. Remove Customer ID from Color.
3. Drag Customer ID next to Product on the Columns shelf.

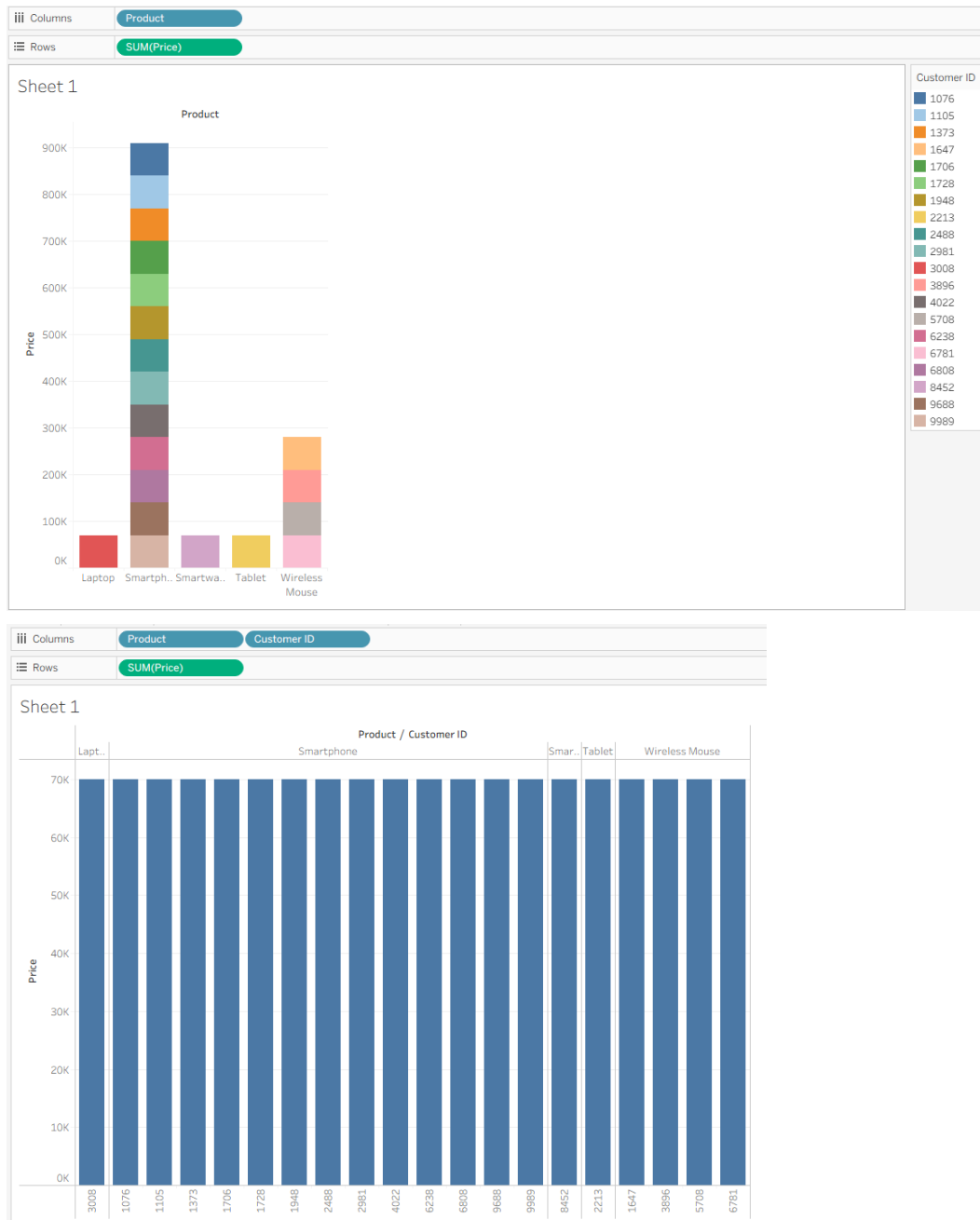
Why we are doing it

A grouped bar chart displays separate bars for each customer under every product.

Usage

Use a grouped bar chart when you want to compare product prices between different customers.

Output:-



b. Build treemaps and pie charts for part-to-whole analysis

Step B: Building Treemaps and Pie Charts

1. Building a Treemap

1. Open a New Worksheet.
2. Drag Product to the workspace.
3. Drag Price to the workspace.
4. Click Show Me.
5. Select Treemap.

Why we are doing it

A treemap shows products as rectangles where larger rectangles represent higher prices.

Usage

Use a treemap to compare the contribution of each product to the total price.

2. Building a Pie Chart

1. Open a New Worksheet.
2. Click Show Me.
3. Select Pie Chart.
4. Drag Product to Color.
5. Drag Price to Angle.
6. Drag Price to Label (optional).

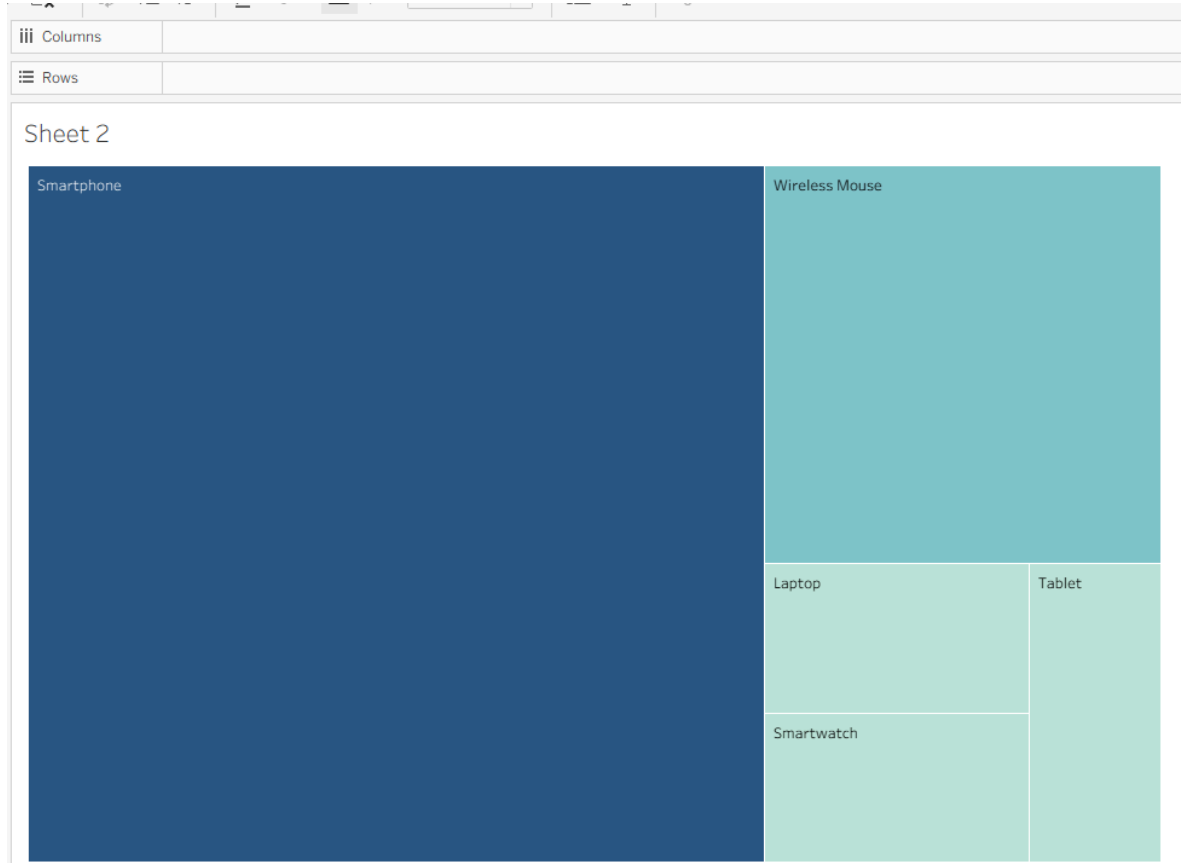
Why we are doing it

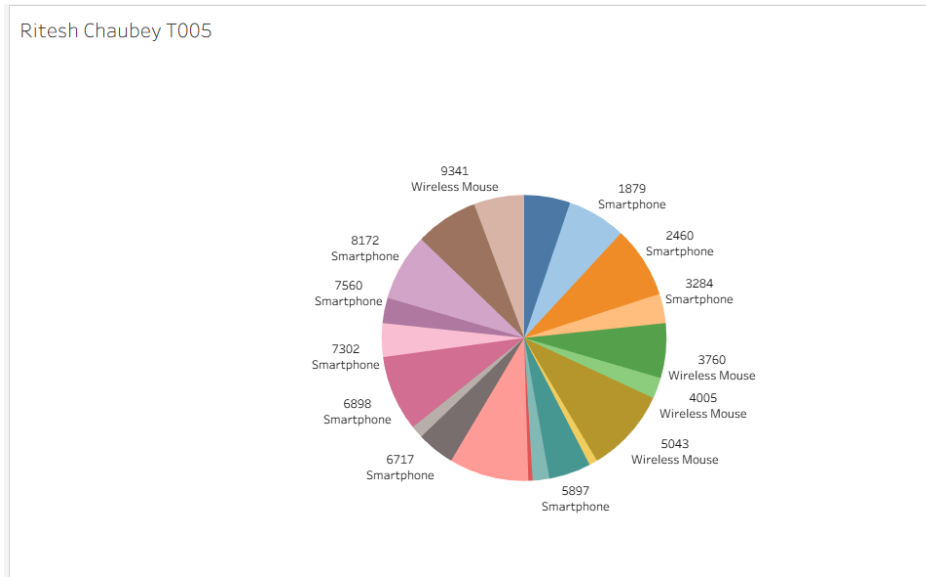
A pie chart displays each product as a slice of the total price.

Usage

Use a pie chart when there are only a few product categories.

Output:-





c. Design scatter plots to identify relationships

Step C: Designing Scatter Plots

1. Building the Scatter Plot

1. Open a New Worksheet.
2. Drag Practical_3(Dataset) (Count) (*green field*) to the Columns shelf.
3. Drag Price to the Rows shelf.
4. Drag Product to Detail.
5. Drag Customer ID to Color.

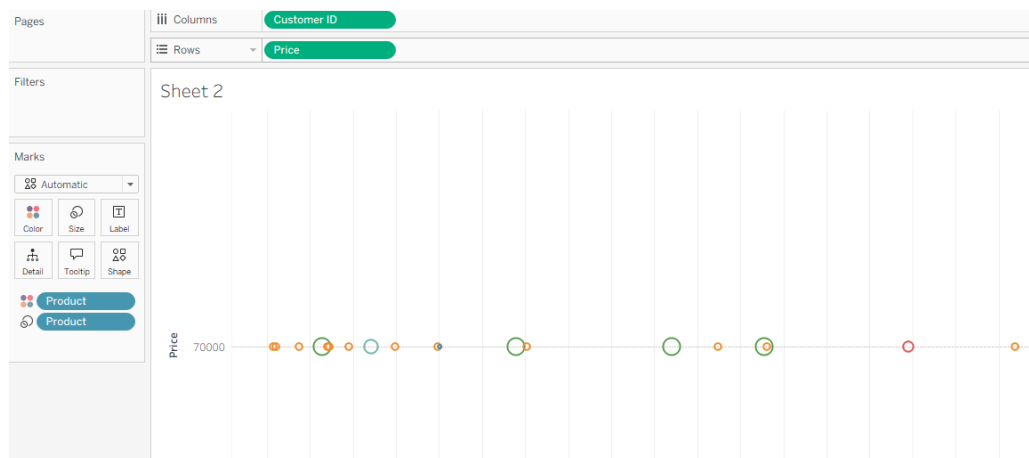
Why we are doing it

A scatter plot shows the relationship between product prices and the number of records.

Usage

Use a scatter plot to identify trends, clusters, and outliers.

OUTPUT:-



d. Create dual-axis and combination charts

Step D: Creating Dual-Axis and Combination Charts

1. Preparing the Fields

1. Convert Date to Date datatype if necessary.

2. Building the Dual-Axis Chart

1. Open a New Worksheet.
2. Drag Date to the Columns shelf.
3. Drag Price to the Rows shelf.
4. Drag Practical_3(Dataset) (Count) to the Rows shelf beside Price.
5. Right-click the second measure and select Dual Axis.
6. In the Marks card:
 - Change Price to Bar.
 - Change Count to Line.

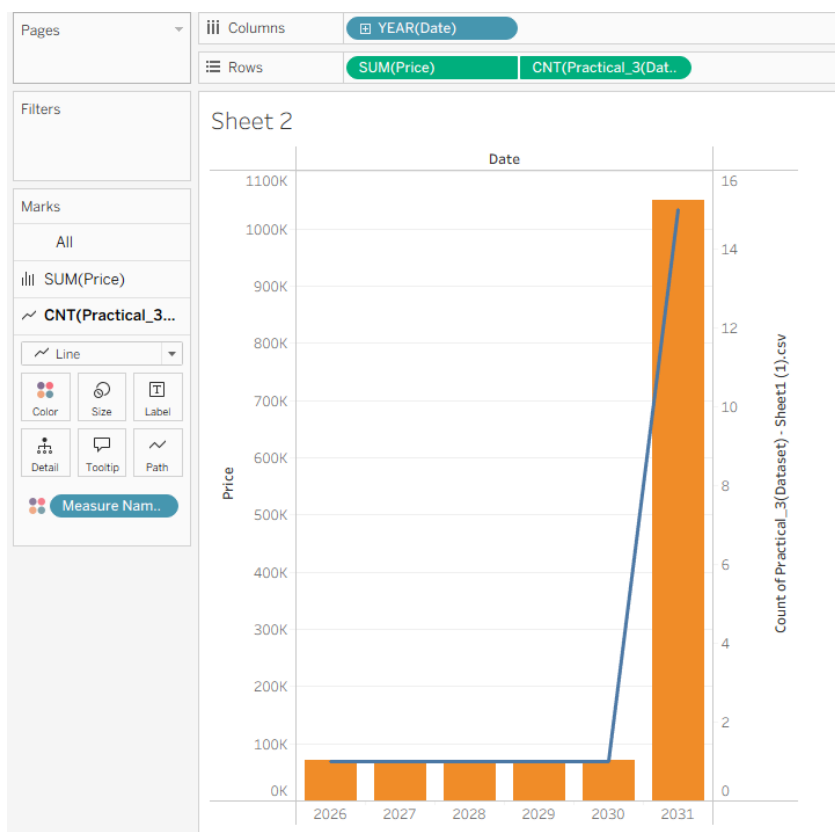
Why we are doing it

A dual-axis chart compares product prices with the number of records over time.

Usage

Use this chart to compare two different measures on the same timeline.

OUTPUT:-



e. Develop histograms for distribution analysis

Step E: Developing Histograms

1. Building the Histogram

1. Open a New Worksheet.

2. Select the Price measure.
3. Click Show Me.
4. Select Histogram.

Tableau automatically creates:

- Price (bin)
- CNT(Price)

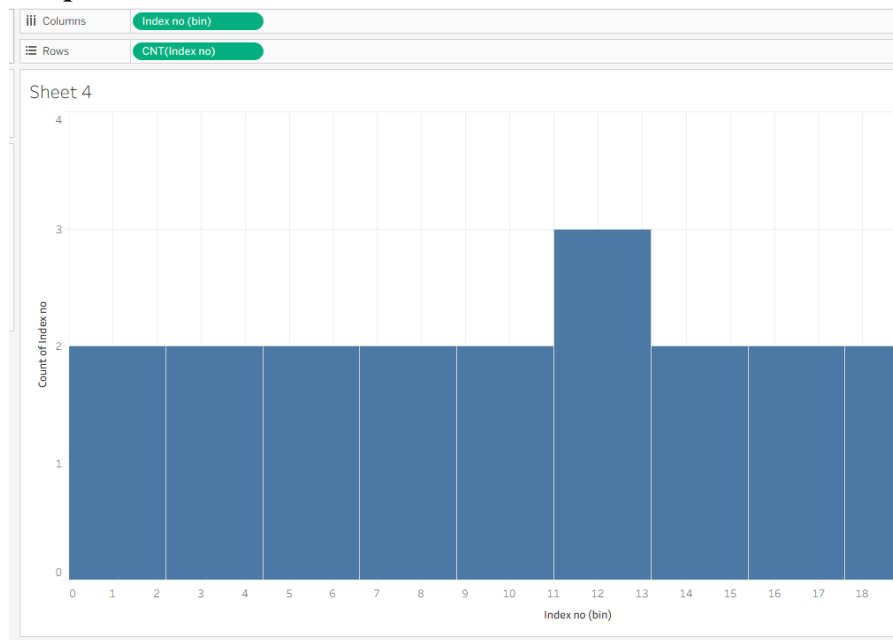
Why we are doing it

A histogram groups prices into ranges and shows how many records fall into each range.

Usage

Use a histogram to understand the distribution of prices.

Output:-



f. Build box-and-whisker plots

Step F: Building Box-and-Whisker Plots

1. Building the Box Plot

1. Open a New Worksheet.
2. Drag Product to the Columns shelf.
3. Drag Price to the Rows shelf.
4. Click Analysis → Uncheck Aggregate Measures.
5. Click Show Me.
6. Select Box-and-Whisker Plot.

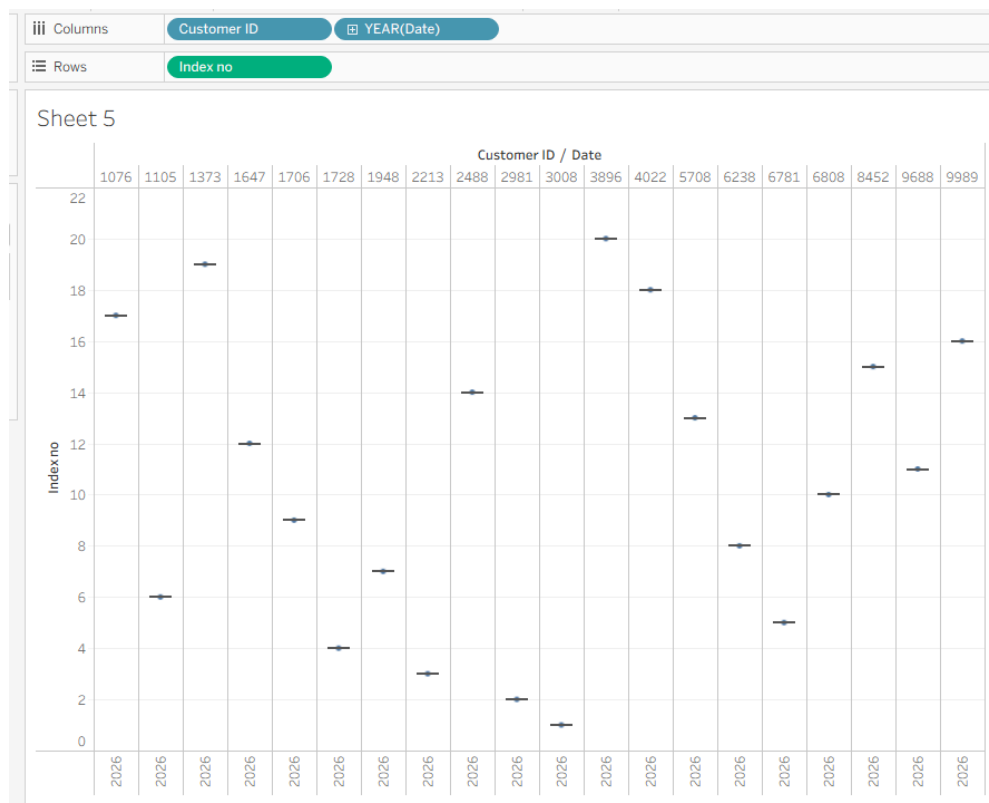
Why we are doing it

A box plot shows the minimum, maximum, median, and spread of prices for each product.

Usage

Use a box plot to compare price distributions and detect outliers.

Output:-



G. Use Gantt charts and advanced date visualizations

Step G: Building Gantt Charts

1. Building the Gantt Chart

1. Open a New Worksheet.
2. Drag Date to the Columns shelf.
3. Tableau will automatically display YEAR(Date).
4. Drag Customer ID to the Rows shelf.
5. In the Marks card, change Automatic to Gantt Bar.
6. Drag Index no to the Size card.
7. Drag Price to the Color card.

Why we are doing it

A Gantt chart displays customer records along a timeline. The Date field represents the timeline, Customer ID identifies each record, Index no controls the length of the Gantt bars, and Price highlights each record using different colors.

Usage

Use a Gantt chart to visualize customer records over time and compare transactions across different years. It helps in understanding the timeline of records while showing price variations through color.

Output:-

